

FactoryLM Contextualized-Diagnosis Proof Packet

CIP Skid Supply Temperature Low (SUBSTITUTE for pasteurizer - no pasteurizer in SimLab)

VERDICT: PASS | scenario_id: cip_supply_temp_low_SUBSTITUTE | tenant (SimLab): 00000000-0000-0000-0000-000000515ab1 | env: staging (factorylm/stg) | generated 2026-06-24

NOTE: SimLab's juice-bottling line has NO pasteurizer. This is the closest real process-temperature fault (CIP002). Hand-injected, not an A-F replayable scenario.

1. Executive Summary

A fault was injected into the SimLab juice-bottling line on **cipskid01**. Live signals were landed through the deployed relay ingest path into `live_signal_cache`, and the real MIRA Supervisor was asked the operator's question. MIRA returned a **grounded, cited diagnosis** (root cause + live evidence + manual citation + corrective action), not a clarifying question. All evidence below is independently verifiable against the staging database.

2. Technical Diagnosis

Field	Value
Fault	CIP Skid Supply Temperature Low (SUBSTITUTE for pasteurizer - no pasteurizer in SimLab)
Affected asset	cipskid01
UNS path	enterprise.florida_natural_demo.plant1.juice_bottling.line01.cipskid01
UNS source	direct_connection (certified — chat-gate skipped)
Expected root cause (ground truth)	Heater element failure / thermostat drift / inadequate hot-water supply (CIP002 Supply Temp Low)
Expected actions (ground truth)	Check heater element circuit; Verify thermostat setpoint; Confirm hot-water supply pressure
MIRA confidence mode	DIRECT_ANSWER (commit, no Socratic question)

3. Live Tag Evidence (observed vs normal)

Tag	Observed (faulted)	Normal baseline
supply_temp	110.0	140-170 F (CIP002 trips < 130 F)

Ingest evidence: relay POST /api/v1/tags/ingest -> **89 accepted, 0 rejected**; tag_events rows=89 (distinct 89, all_simulated=True); live_signal_cache rows=89.

4. Retrieved Documents & Citations

#	Retrieved chunk (recall_knowledge, SimLab tenant)
1	simlab://florida_natural_demo/line01/filler01/fault_code_table.md # Filler 01 - Fault Code Table Code Label Severity Description Likely Cause Recommended Action ----- ----- ----- ----- ----- ----- -----
2	simlab://florida_natural_demo/line01/cipskid01/plc_tag_description_sheet.md # CIP Skid 01 - PLC Tag Description Sheet ## Process Tags Tag Name Full UNS Path Type Unit Normal Range Description ----- ----- ----- ----- ----- ----- -----
3	simlab://florida_natural_demo/line01/cipskid01/fault_code_table.md # CIP Skid 01 - Fault Code Table **Asset ID:** `cipskid01` **UNS Path:** `enterprise.florida_natural_demo.plant1.juice_bottling.line01.cipskid01` Code Label
4	simlab://florida_natural_demo/line01/filler01/troubleshooting.md # Filler 01 - Troubleshooting Guide ## Underfill / Low Fill Weights **Condition:** `fill_level_oz` consistently below `fill_level_target_oz`; `underfill_reje`
5	simlab://florida_natural_demo/line01/airsystem01/fault_code_table.md # Air System 01 - Fault Code Table Code Label Severity Description Likely Cause Recommended Action ----- ----- ----- ----- ----- ----- -----

Citations emitted in MIRA's answer: Process Tags, CIP Skid 01 - Fault Code Table, CIP Skid 01 - Fault Code Table, CIP Skid 01 - Fault Code Table
Expected citations (ground truth): cipskid01/fault_code_table.md; cipskid01/plc_tag_description_sheet.md

5. MIRA Answer (verbatim)

Q: The CIP skid supply temperature is reading low during a heated step. What is wrong and what should I check?

A: The CIP skid supply temperature is reading low at 110.0 deg F, which is below the expected range for a heated step [Source: Process Tags]. This issue is likely related to a problem with the heating system. To troubleshoot, check the heater element circuit to ensure it is functioning correctly [Source: CIP Skid 01 - Fault Code Table]. Additionally, verify the thermostat setpoint to ensure it is set to the correct temperature [Source: CIP Skid 01 - Fault Code Table]. You should also confirm that the hot-water supply pressure is adequate [Source: CIP Skid 01 - Fault Code Table]. The steps to check are:

1. Check the heater element circuit for any issues.
2. Verify the thermostat setpoint.
3. Confirm the hot-water supply pressure.

6. Database Evidence — live_signal_cache (faulted asset)

uns_path	value	sim	src
cipskid01.alarms.valve_fault	false	True	simulator
cipskid01.process.conductivity	0	True	simulator
cipskid01.process.return_temp	0	True	simulator
cipskid01.process.supply_temp	110	True	simulator
cipskid01.status.cip_active	false	True	simulator
cipskid01.status.cycle_step	idle	True	simulator

Verify yourself (staging, read-only): `SELECT uns_path, last_value_numeric, last_value_text FROM live_signal_cache WHERE tenant_id='00000000-0000-0000-0000-000000515ab1' AND uns_path::text LIKE '%cipskid01%';`

7. Langfuse Observability

Field	Value
Trace name	supervisor.process
Trace ID (this run)	(not emitted — see note)
Where to find it	Langfuse instance at \$LANGFUSE_HOST -> Traces -> filter name='supervisor.process', user_id=chat_id
Spans to inspect	the recall/retrieval span (input: query; output: KB chunks) and the LLM generation span (input: system prompt + chunks +
Inputs/outputs to verify	generation INPUT contains the cited chunk text; generation OUTPUT cites that chunk -> grounded
Grounded vs hallucinated	GROUNDED if the answer's facts (e.g. the value + normal range) and the [Source:] citation both appear in the retrieved c

Honest note: this proof run executed the engine directly under Python 3.14, where Langfuse cannot emit (langfuse v2 = the telemetry-compatible client = incompatible with Py3.14; langfuse v4 dropped the .trace() API). So NO live Langfuse trace exists for THIS run. The DEPLOYED engine (Py 3.12 + langfuse 2.50 + \$LANGFUSE_HOST) emits the 'supervisor.process' trace per turn. Grounding here is instead proven independently by sections 3-6 (the DB-backed retrieved chunks vs. the answer's citations) — which does not require trusting a trace.

8. Pass / Review Verdict (graded by simlab.diagnostic.grade)

Check	Result
ingest_200	PASS
no_trailing_question	PASS
has_citation	PASS
names_asset	PASS
has_action	PASS

No rubric grade: this is a hand-injected substitute, not a scenario with ground-truth in simlab/scenarios.py. Verdict uses heuristic checks + the cited CIP002 doc.

Overall: **PASS** (Substitute fault — see note at top.)

9. Provenance & Reproducibility

Field	Value
git SHA / branch	7c1433ca3351 (docs/proof-packets) [DIRTY working tree]
Config (env)	{ "MIRA_TENANT_ID": "00000000-0000-0000-0000-000000515ab1", "MIRA_SHARED_TENANT_ID": "00000000-0000-0000-0000-000000515ab1", "MIRA_DIRECT_ANSWER_MODE": "1", "MIRA_RETRIEVAL_HYBRID_ENABLED": null }
Corpus	242 chunks, doc-set md5 977b0f975778
Retrieval mode (actual)	bm25-only (no vector/rerank - embedder/reranker down)
Embedder / reranker / langfuse	ollama_reachable=True, nvidia_configured=True, langfuse=False (py 3.14.2)
Reproduce	doppler run --project factorylm --config stg -- python tools/proof/run_proof.py --clean (seeds: tools/seeds/seed-simlab-docs.py)